

Prolog Program for Forward Chaining

1. Aim

To write and execute a Prolog program that implements forward chaining inference, deriving new facts from known facts and rules until no more facts can be derived, and to incorporate the required queries.

2. Objective

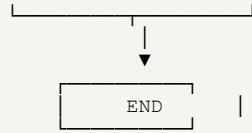
- To understand the forward chaining strategy of inference (data-driven reasoning).
- To represent known facts and if-then rules separately in Prolog.
- To implement an inference loop that repeatedly applies rules to derive new facts.
- To retrieve and display all facts derived through forward chaining.

3. Algorithm

1. Start the program.
2. Create a Prolog database of known facts using the predicate fact/1.
3. Create a set of if-then rules using the predicate rule/2, each linking a condition to a conclusion.
4. Define forward_chain/0: find a rule whose condition is a known fact and whose conclusion is not yet known.
5. Assert the new conclusion as a fact and print it.
6. Repeat the process until no new facts can be derived.
7. Enter a query to check whether a target fact has been derived.
8. Display whether the target fact is now known.
9. Stop the program.

4. Flowchart





5. Code with Output

Code:

```
% -----  
% Known facts  
% -----  
:- dynamic(fact/1).  
fact(has_feathers(tweety)).  
fact(lays_eggs(tweety)).  
  
% -----  
% Rules: condition -> conclusion  
% -----  
rule(has_feathers(X), bird(X)).  
rule(bird(X), can_fly(X)).  
rule(lays_eggs(X), oviparous(X)).  
  
% -----  
% Forward chaining engine  
% -----  
forward_chain :-  
    rule(Cond, Concl),  
    fact(Cond),  
    \+ fact(Concl),  
    assertz(fact(Concl)),  
    format("Derived: ~w~n", [Concl]),  
    forward_chain.  
forward_chain.
```

Output:

```
?- forward_chain.  
Derived: bird(tweety)  
Derived: can_fly(tweety)  
Derived: oviparous(tweety)  
true.  
  
?- fact(can_fly(tweety)).  
true.
```

6. Result

The Prolog program was successfully implemented to perform forward chaining, repeatedly applying rules to known facts until all derivable conclusions (bird, can_fly, oviparous) were added to the fact base.

7. Conclusion

Thus, forward chaining was successfully implemented using Prolog. The experiment demonstrates data-driven inference: starting from known facts, rules are fired whenever their conditions are satisfied, and assertz/1 grows the fact base until a fixpoint is reached.